Chapter 1 Study Notes: Foundations & Evolution of Deep Learning

1. Resolving Core Confusions - Comparison Reference

Common Industry Misunderstandings

Concept A	Concept B	Key Distinctions
Machine Learning	Deep Learning	Machine learning is a broad discipline where a program is defined by learning from data rather than being entirely hard-coded. Deep Learning is a specialized subset of machine learning that specifically uses neural networks containing multiple layers.
Architecture	Model	Architecture is the general template or mathematical function form that we fit. A model is the specific combination of that architecture with a particular, finalized set of parameters.
Loss	Metric	Loss is a measure of performance used to guide training and update parameters automatically via optimization. A metric is a measurment of model quality calculated on the validation set strictly for human consumption and interpretation.
Classification	Regression	Classification models attempt to predict a discrete category or class from distinct possibilities (e.g. "dog" or "cat"). Regression models attempt to predict one or more continous numeric quantities (e.g. a temperature or location)